



brms Bayesian Multilevel Models Using Stan

A source-faithful paper talk from Buerkner 2017

Research Presentation Program

Department of Statistics & Data Science

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香港中文大學
The Chinese University of Hong Kong

brms exposes multilevel covariance as Stan-ready parameters

$$\mathbf{V}_k = \mathbf{D}(\boldsymbol{\sigma}_k) \boldsymbol{\Omega}_k \mathbf{D}(\boldsymbol{\sigma}_k), \quad \boldsymbol{\Omega}_k \sim \text{LKJ}(\zeta), \quad \boldsymbol{\Sigma}_k = \mathbf{V}_k \otimes \mathbf{A}_k$$

Buerkner uses this parameterization to let brms express group-level standard deviations, correlations, and known covariance structures before Stan/HMC fitting.

Model components

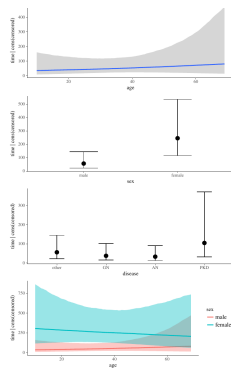
- group-level covariance matrix
- standard-deviation vector
- LKJ correlation prior

Interpretation

The paper's central object is a multilevel formula interface whose group effects map to covariance parameters and LKJ correlation priors.

Source: Buerkner 2017, Section 2.1, DOI 10.18637/jss.v080.i01.

The kidney example turns posterior draws into interpretable effects

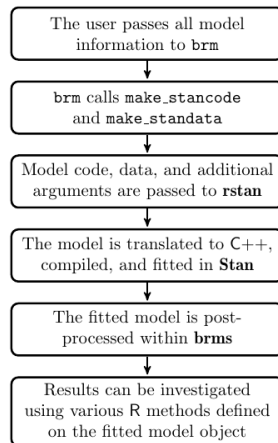


1: Marginal effects plots of all population-level predictors of the kidney model drawn on 4.

The result page stays on the paper's kidney example rather than importing the Stage 4 clustered-calibration fixture.

Figure 3 crop, Buerkner 2017, CC BY.

The production path in the paper is formula to Stan to brmsfit



description of the model fitting proc

The diagram is from the paper: brms handles Stan-code generation and post-processing while Stan performs the compiled Bayesian fit.

The comparison table is broad, but not a blank check

	brms	lme4	MCMCglmm
<i>Supported model types:</i>			
Linear models	yes	yes	yes
Robust linear models	yes	no	no
Binomial models	yes	yes	yes
Categorical models	yes	no	yes
Multinomial models	no	no	yes
Count data models	yes	yes	yes
Survival models	yes ¹	yes	yes
Ordinal models	various	no	cumulative
Zero-inflated and hurdle models	yes	no	yes
Generalized additive models	yes	no	no
Non-linear models	yes	no	no
<i>Additional modeling options:</i>			
Variable link functions	various	various	no
Weights	yes	yes	no
Offset	yes	yes	using priors
Multivariate responses	limited	no	yes
Autocorrelation effects	yes	no	no
Category-specific effects	yes	no	no
Standard errors for meta-analysis	yes	no	yes
Censored data	yes	no	yes
Truncated data	yes	no	no
Customized covariances	yes	no	yes
Missing value imputation	no	no	no
<i>Bayesian specifics:</i>			
Parallelization	yes	–	no
Population-level priors	flexible	–.3	normal
Group-level priors	normal	–.3	normal
Covariance priors	flexible	–.3	restricted ¹
<i>Other:</i>			
Estimator	HMC, NUTS	ML, REML	MH, Gibbs ²
Information criterion	WAIC, LOO	AIC, BIC	DIC
C++ compiler required	yes	no	no
Modularized	no	yes	no

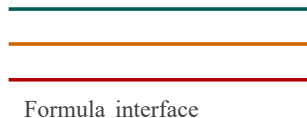
Table 1: Comparison of the capabilities of the **brms**, **lme4** and **MCMCglmm** packages. Notes: (1) Weibull family only available in **brms**. (2) Estimator consists of a combination of both

The table supports a balanced interpretation: broad formula coverage, but Bayesian modelling choices and some features remain outside automatic handling. Table 1 crop, Buerkner 2017, CC BY.

The paper's practical rule is interface first, diagnostics always

Failure evidence

A syntactically valid model is not automatically a meaningful Bayesian analysis.



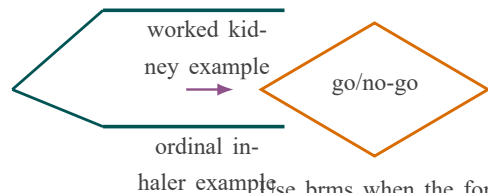
motivates

Manipulate sampling

- choose priors deliberately
- inspect chains and posterior plots
- compare with WAIC/LOO

Comparator arms

Decision rule



Use brms when the formula path is faithful, then audit priors, diagnostics, and predictive comparison.

The talk closes on the paper's modelling contract: brms automates Stan plumbing, not scientific judgement.